

B.Tech: CSE (BTCS) Last Year

Year: 2025-26

Deep Learning & Neural Network Lab

Enrolment No.: MITU22BTCS0153

Division: LY CORE 3

Roll No.: 2223263

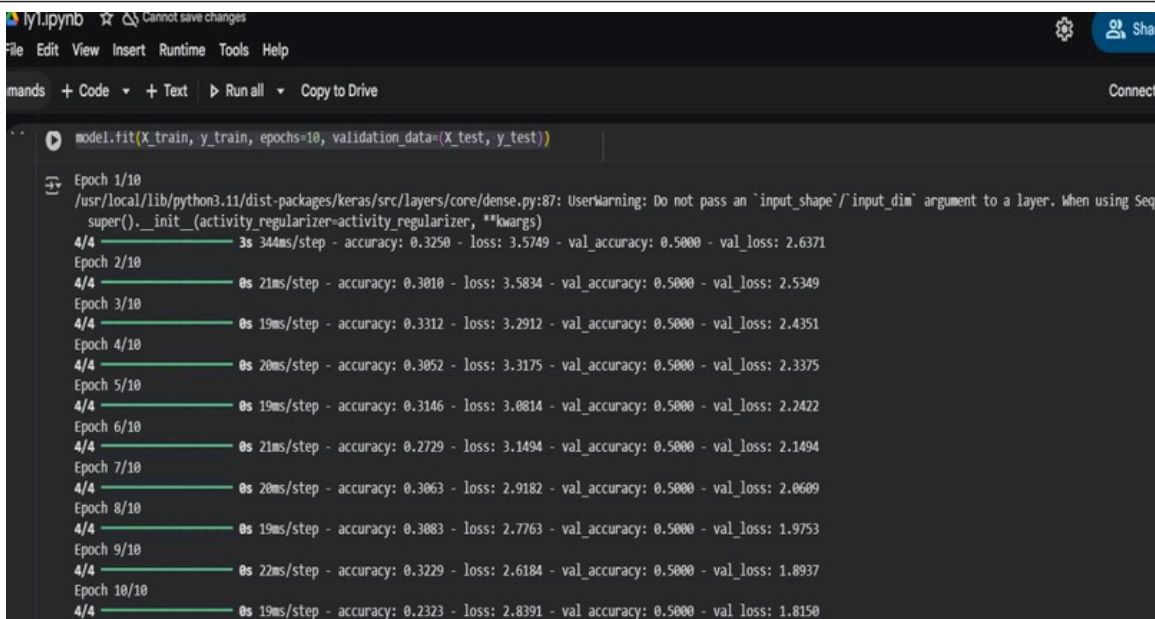
Name: Arya Deshmukh

Experiment No 01

Aim:	Introduction to Keras and Tensorflow (Optional – Pytorch). Configure and use google colab and kaggle GPU
Objectives:	1. To configure anaconda and google colab, kaggle environment 2. To Explore TF/Keras/Pytorch libraries 3. To learn to use GPU/TPU 4. To learn and understand Git TensorFlow is an open-source deep learning framework developed by Google Brain. It is designed for numerical computation and large-scale machine learning. It provides a flexible ecosystem of tools and libraries to build and deploy ML models across various platforms (CPUs, GPUs, and TPUs).
Theory	TensorFlow is an open-source deep learning framework developed by Google Brain. It is designed for numerical computation and large-scale machine learning. It provides a flexible ecosystem of tools and libraries to build and deploy ML models across various platforms (CPUs, GPUs, and TPUs). Key Features of TensorFlow: • Supports both low-level operations (with tensors, sessions, and graphs) and high-level APIs (like Keras). • Enables automatic differentiation using the computation graph. • Optimized for both training and inference on multiple hardware devices. • Provides TensorBoard for visualization and debugging. • Supports distributed training and deployment using TensorFlow Serving and TensorFlow Lite. TensorFlow is an open-source deep learning framework developed by Google Brain. It is designed for numerical computation and large-scale machine learning. It provides a flexible ecosystem of tools and libraries to build and deploy ML models across various platforms (CPUs, GPUs, and TPUs). Key Features of TensorFlow: • Supports both low-level operations (with tensors, sessions, and graphs) and high-level APIs (like Keras). • Enables automatic differentiation using the computation graph. • Optimized for both training and inference on multiple hardware devices.

	• Provides TensorBoard for visualization and debugging. • Supports distributed training and deployment using TensorFlow Serving and TensorFlow Lite. PyTorch is another popular open-source deep learning framework developed by Facebook's AI Research Lab (FAIR). It is known for dynamic computation graphs and easy debugging. Features of PyTorch: • Uses eager execution, which makes debugging easier. • Provides torch.nn for building neural networks. • Offers GPU acceleration using CUDA. • Widely used for research and production in NLP and Computer Vision.
Code	<pre>import tensorflow as tf print("GPU Available:",tf.config.list_physical_devices('GPU')) import tensorflow as tf from tensorflow import keras from tensorflow.keras import layers,models from tensorflow.keras.models import Sequential from tensorflow.keras.layers import Dense from sklearn.datasets import load_iris from sklearn.model_selection import train_test_split from sklearn.preprocessing import LabelBinarizer #Load and prepare data data = load_iris() X= data.data y= LabelBinarizer().fit_transform(data.target) X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2) model = Sequential() model.add(Dense(10, input_shape=(4,), activation='relu')) model.add(Dense(3, activation='softmax')) model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy']) model.fit(X_train, y_train, epochs=10, validation_data=(X_test, y_test))</pre>

Output (Paste)	GPU Available: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
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Screenshot)	 <pre> model.fit(X_train, y_train, epochs=10, validation_data=(X_test, y_test)) Epoch 1/10 /usr/local/lib/python3.11/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape` / `input_dim` argument to a layer. When using Sequential().__init__(activity_regularizer=activity_regularizer, **kwargs) 4/4 — 3s 344ms/step - accuracy: 0.3250 - loss: 3.5749 - val_accuracy: 0.5000 - val_loss: 2.6371 Epoch 2/10 4/4 — 0s 21ms/step - accuracy: 0.3010 - loss: 3.5834 - val_accuracy: 0.5000 - val_loss: 2.5349 Epoch 3/10 4/4 — 0s 19ms/step - accuracy: 0.3312 - loss: 3.2912 - val_accuracy: 0.5000 - val_loss: 2.4351 Epoch 4/10 4/4 — 0s 20ms/step - accuracy: 0.3052 - loss: 3.3175 - val_accuracy: 0.5000 - val_loss: 2.3375 Epoch 5/10 4/4 — 0s 19ms/step - accuracy: 0.3146 - loss: 3.0814 - val_accuracy: 0.5000 - val_loss: 2.2422 Epoch 6/10 4/4 — 0s 21ms/step - accuracy: 0.2729 - loss: 3.1494 - val_accuracy: 0.5000 - val_loss: 2.1494 Epoch 7/10 4/4 — 0s 20ms/step - accuracy: 0.3063 - loss: 2.9182 - val_accuracy: 0.5000 - val_loss: 2.0609 Epoch 8/10 4/4 — 0s 19ms/step - accuracy: 0.3083 - loss: 2.7763 - val_accuracy: 0.5000 - val_loss: 1.9753 Epoch 9/10 4/4 — 0s 22ms/step - accuracy: 0.3229 - loss: 2.6184 - val_accuracy: 0.5000 - val_loss: 1.8937 Epoch 10/10 4/4 — 0s 19ms/step - accuracy: 0.2323 - loss: 2.8391 - val_accuracy: 0.5000 - val_loss: 1.8150 </pre>
Conclusion :	This experiment provided an understanding of deep learning frameworks like TensorFlow, Keras, and PyTorch, and how to use Google Colab and Kaggle GPUs for faster model training. It demonstrated how GPU acceleration enhances the performance of neural network computations and simplifies running deep learning models efficiently in cloud environments.

Grade / Marks:	Subject Teacher Signature:
	Date: